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15 JULY 1959
SUPERSEDING
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10 APRIL 1956

# MILITARY SPECIFICATION

## PLOTTING BOARD, TACTICAL DISPLAY, PT-429/A

This specification has been approved by the Bureau of Aeronautics, Department of the Navy

## 1. SCOPE

1.1 Scope - The equipment covered by this specification shall be capable of receiving inputs from a Ground Position Indicator in the form of signals representing North-South and East-West ground miles traveled and True Heading and shall be capable of displaying this information visually.

1.2 Classification - The Plotting Board, Tactical Display, PT-429/A covered by this specification shall be of one type and shall consist of the following:

| Unit | Type Designation | Max. Allow. Size (Inches) H | W | D | Max. Allow. Wt. (Lbs) | Appl. Par. |
|---|---|---|---|---|---|---|
| Plotting Board, Tactical Display | PT-429/A | 11 | 10.5 | 3.06 | 12.5 | 3.4.1 |

## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-17555 | Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts |
| MIL-E-4682 | Electron Tubes, Choice and Application of |
| MIL-W-5088 | Wiring; Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |

FSC 6605



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| MIL-I-6181 | Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment |
| MIL-E-7894 | Electric Power, Aircraft Characteristics of |
| MIL-D-18300 | Design Data Requirements for Contracts Covering Airborne Electronics Equipment |
| MIL-T-18303 | Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Airborne Electronic and Associated Equipment |

STANDARDS

Military

| | |
|---|---|
| MS 25245 | Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 115/200 Volt 3-Phase AC, or Combinations |

PUBLICATIONS

Bureau of Aeronautics

| | |
|---|---|
| SAR-300 | Material Changes and Material Bulletins, Preparation of |

2.2 Availability of Documents:

(1) When requesting applicable documents give both the title and number. All requests shall be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), Philadelphia 11, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:



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(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event that the entire assembly is built to be expended, the assembly shall be so constructed and the quality of the individual parts shall be of such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

(5) Nonmagnetic materials shall be used for all parts, except where magnetic materials are required.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400, except as follows:

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Nonstandard parts of an electrical nature as defined in Specification MIL-E-5400, shall be approved before the preproduction model of the equipment is submitted for tests. The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Nonstandard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

* 3.1.2 Tube Types - The choice and application of electron tubes and semiconductor devices used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube or semiconductor device not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information.

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

* 3.1.2.1 Electron Tube Complement - A tube complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.



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* 3.2 Design and Construction:

* 3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

* 3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 12-1/2 pounds.

* 3.2.3 Operating Life -

* 3.2.3.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

* 3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4 Cabling and Connections -

* 3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. All electrical connections to the PT-429/A shall be made through a single MS standard connector.

* 3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

* 3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

* 3.2.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. A durable nameplate shall be securely attached to the exterior of the PT-429/A. All markings shall be legible and durable.

* 3.2.6 Interference Control - The generation of radio interference by the equipment shall be controlled within the limits of Specification MIL-I-6181.

* 3.2.7 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

* 3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient ($30^{\circ} \pm 10^{\circ}$C) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative humidity |
| Input power voltage | 115 ±1.0V AC and/or 27.5 ±0.5 V DC |

3.2.9 Service Conditions - The plotting board shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment except it shall not be required to operate above 15,000 feet altitude.

* 3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits as specified hereinafter. The total power required shall not exceed 100 watts.



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(1) AC Power (Single Phase)

115-Volt Constant Frequency (Type I)

| | |
|---|---|
| Operating voltage limits | 105 to 125 volts |
| Operating frequency limits | 360 to 440 cps |
| Reduced performance voltage limits | (see paragraph 3.2.7) |

(2) DC Power 28-Volt System

| | |
|---|---|
| Operating voltage limits | 24 to 30 volts |

* 3.2.11 AC Limits - There are no requirements for performance at frequencies below 360 cps. Down to 200 cycles the voltage shall be considered as maintained within the operating limits. Below 200 cycles, the voltage shall be considered as varying proportionally with the frequency, with zero voltage existing at zero frequency.

3.2.12 Power Input Circuit - The negative side of the d-c power line shall be internally grounded to the frame of the indicator. The 115-volt 400-cycle power input terminals to the PT-429/A shall be isolated from the frame to the PT-429/A.

* 3.3 Performance

* 3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under standard conditions. When reduced performance under environmental test conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified. The terms used throughout the remainder of this specification are defined as follows:

"Bug" refers to the carriage which supports the lamp and lens system and the projected spot of light representing the position of the aircraft.

Track refers to motion of the bug across the screen when simulating the motion of the aircraft.

Slew refers to the process whereby the bug is moved to a desired spot on the screen by use of appropriate controls.

Range Scale (abbreviated SCALE on the front panel) refers to the number of nautical miles represented by the diameter of the screen.

Mode refers to indicator condition of operation.

3.3.2 Signal Inputs - The bug shall be positioned by 2 synchro signals obtained from Eclipse-Pioneer Type AY/201-4 synchros or equivalent, one signal shall represent North-South distance traveled at a scale factor of 1/2 revolution per nautical mile and the other signal will represent East-West distance traveled at the same scale factor. The arrow on the bug shall be oriented by a signal representing the heading of the aircraft obtained from an Eclipse-Pioneer type AY-201-4 synchro or equivalent; one revolution of the arrow shall represent 360° in azimuth. The indicator shall provide 26 volts ac 1/4 amp. excitation at line frequency for the heading synchro transmitter.

* 3.3.3 Bug Follow-up Speed and Accuracy - The bug shall follow input signals representing 200 knots on the 10 mile range and 400 knots on the 25 and 100 mile ranges. The accuracy shall be within 1/32 inch.

* 3.3.4 Accuracy of True Heading Arrow - The accuracy shall be within ± 1.5 degrees.



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3.4 Detail Requirements -

* 3.4.1 Plotting Board, Tactical Display PT-429/A - This equipment shall meet the following detail requirements.

* 3.4.1.1 Function - The function of the PT-429/A equipment is to plot and display input data representing aircraft ground miles traveled and aircraft true heading.

* 3.4.1.2 Form Factor and Mounting - Refer to figure 1 for data on dimensions and mounting. The PT-429/A shall be provided with a suitable mounting pad capable of supporting the weight of the instrument and accessible from the rear of the indicator. No external shock mounts are required.

* 3.4.1.3 Weight - The unit weight is 12 lbs 8 oz., excluding cables.

3.4.1.4 Mechanical Design - The instrument shall be designed so that the various functions required by this specification and indicated in the block diagram (figure 2) are accomplished. The PT-429/A shall meet all of the requirements of MIL-E-5400 except as noted herein.

3.4.1.5 Front Panel - The general layout of the front panel of the indicator shall conform to figure 1. The motion of the aircraft shall be indicated as a moving spot of light projected on a screen.

3.4.1.6 Screen - The size, markings and general construction of the screen shall conform to figure 1.

3.4.1.7 Screen Lighting - Low level red lighting shall be provided for the purpose of reading the scale graduations along the bearing line, grid lines, compass rose and any marks that may be made on the front surface of the screen. In order to adjust the intensity of the screen lighting, a control marked ILLUMINATION SCREEN, shall be provided on the front panel of the PT-429/A. At 28 volts d.c. the brightness of the screen shall be between 0.4 and 3.0 ft. lamberts.

3.4.1.8 Overlay Mounting Provision - Spring clips shall be provided for holding 8-7/8" diameter by .010" thick transparent plastic overlays on the top surface of the screen. The clips shall not protrude above the front surface of the panel.

3.4.1.9 Bug - The bug shall project a 1/4" diameter spot of light on the top surface of the matte finished rotatable grid disk (see figure 1). The spot of light shall contain an arrow representing the heading of the aircraft. The oriented arrow shall follow a heading input signal when the input signal is changed at a rate of 3 rpm. The shank of the arrow shall contain a cross mark to locate the center of the bug.

3.4.1.10 Bug Lighting - The brightness of the bug shall be adjustable by means of a control, marked ILLUMINATION-BUG, located on the front panel of the PT-429/A. The maximum intensity of the light shall be sufficient to make the bug clearly visible in direct sunlight, but in no case shall it be less than 250 ft. lamberts. Provisions shall be made for convenient replacement of the lamp used as a source of light for the bug.

3.4.1.11 Bug Travel - The drive mechanism for the bug shall be capable of driving the center of the bug anywhere within an 8-1/4" diameter circle concentric with the center of the screen. No damage shall occur to the indicator if the input signals are allowed to override the limits of travel of the bug.

3.4.1.12 Bug Drive Mechanism - The bug shall be driven in North-South and East-West directions by means of lead screws as indicated in figure 2.

3.4.1.13 Controls - The front panel of the indicator shall contain the following controls located as shown in figure 1. All markings on the front panel shall be illuminated with red light.

3.4.1.14 Mode Selector - A control knob marked MODE shall be provided for starting and stopping the motion of the bug. When the mode control knob is placed in the central position marked "STBY", the bug shall stop tracking and hold its present position. When the knob is placed in the track position marked "TR", the bug will immediately begin to track the aircraft motion. The mode control



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knob shall also contain a centering "CEN" position adjacent to the "STBY" position, for automatically slewing the bug to the center of the screen. When the knob is moved momentarily to this position, the bug shall travel to within .020 of an inch of the center of the screen at a rate of not less than 0.8 of an inch per second. When the knob is released, it shall return to the "STBY" position and the bug shall remain in the center position until slewed to a new position or until the mode knob is turned to the track position.

3.4.1.15 Slew Switch - A slewing control, marked SLEW, shall be provided on the front panel of the PT-429/A for positioning the bug on the screen. The control switch shall be a momentary contact type switch with a neutral position. The switch shall be employed in the slew motor circuits so that control of the bug motion can be exercised in the North-South and East-West directions separately or together by moving the switch handle in the desired direction. The bug shall slew at a rate not less than 0.8 of an inch per second. The damping of the slewing mechanism and sensitivity of the SLEW switch shall be such that the bug may be accurately positioned. A guard shall be provided to prevent inadvertent movement of switch handle.

* 3.4.1.16 Scale Selector - A control marked SCALE shall be provided on the front panel of the PT-429/A for selecting the desired scale factor, i.e., ground distance representing per inch of bug travel. Scale factors corresponding to the three scales shall be as follows:

| Scale | Scale Factor |
|---|---|
| 10 | 1-1/4 nautical miles per 1 inch |
| 25 | 3-1/8 nautical miles per 1 inch |
| 100 | 12-1/2 nautical miles per 1 inch |

In addition, the SCALE selector shall have an "OFF" position for controlling AC and DC power inputs to the PT-429/A. When the SCALE selector position is changed from one scale to another the bug will immediately track with the input signals at a rate that is in accordance with the corresponding scale factor.

3.4.1.17 Bearing Line Control - A control, marked BEARING, shall be provided for rotating the grid disk through 360°. The ratio between the control and the disk shall be approximately 7 to 1.

3.4.1.18 Servo Amplifiers - All servo amplifiers shall be completely transistorized and shall contain no vacuum tubes or magnetic amplifiers.

3.4.1.19 Lubrication - Only lubrication which will not adversely affect PT-429/A performance, when operated within the temperature limits specified herein, shall be used. In addition, this lubrication shall not cause deterioration of material.

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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting those tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.



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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration (10 minutes) | Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's. |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.



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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 10 | 1 |
| 11 to 100 | 2 |
| 101 to 200 | 1 |
| Over 200 | 1 for each additional 200 or fraction thereof |

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests



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(2) Vibration Test (90 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 25 | 1 |
| 26 to 200 | 1 |
| 201 to 500 | 1 |
| Over 500 | 1 for each additional 500 or fraction thereof |

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

| | |
|---|---|
| Temperature | Normal room temperature |
| Altitude | Normal room |
| Humidity | Room ambient |
| A.C. Voltage | 115 ± 5 volts (at lowest applicable frequency) |
| D.C. Voltage | 26.5 ± 2.5 volts |

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number



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(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.



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## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

5.2 Unit Packaging - Unit packages shall contain one each PT-429/A, "Handbook Operation and Maintenance Instructions."

## 6. NOTES

6.1 Use - The PT-429/A is intended to be used by a pilot to navigate his aircraft during ASW maneuvers.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Ordering Data - Requisitions, contracts and orders shall specify the specification number, the specific name of the item of equipment, the quantity of each item and the class of packaging required. (See Para. 5.)

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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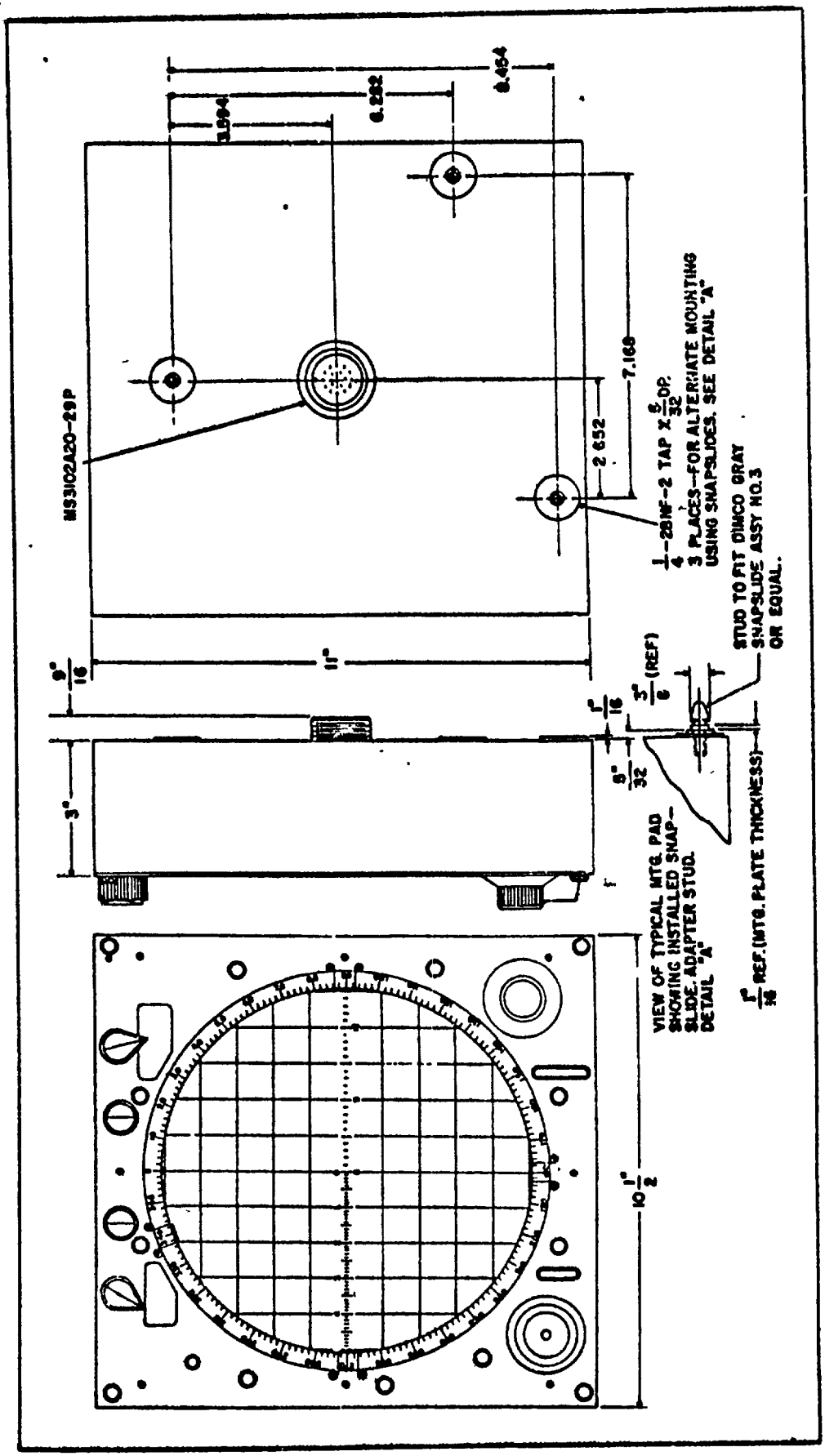



Figure 1. Outline and Installation Dimensions, Plotting Board, Tactical Display, PT-429/A



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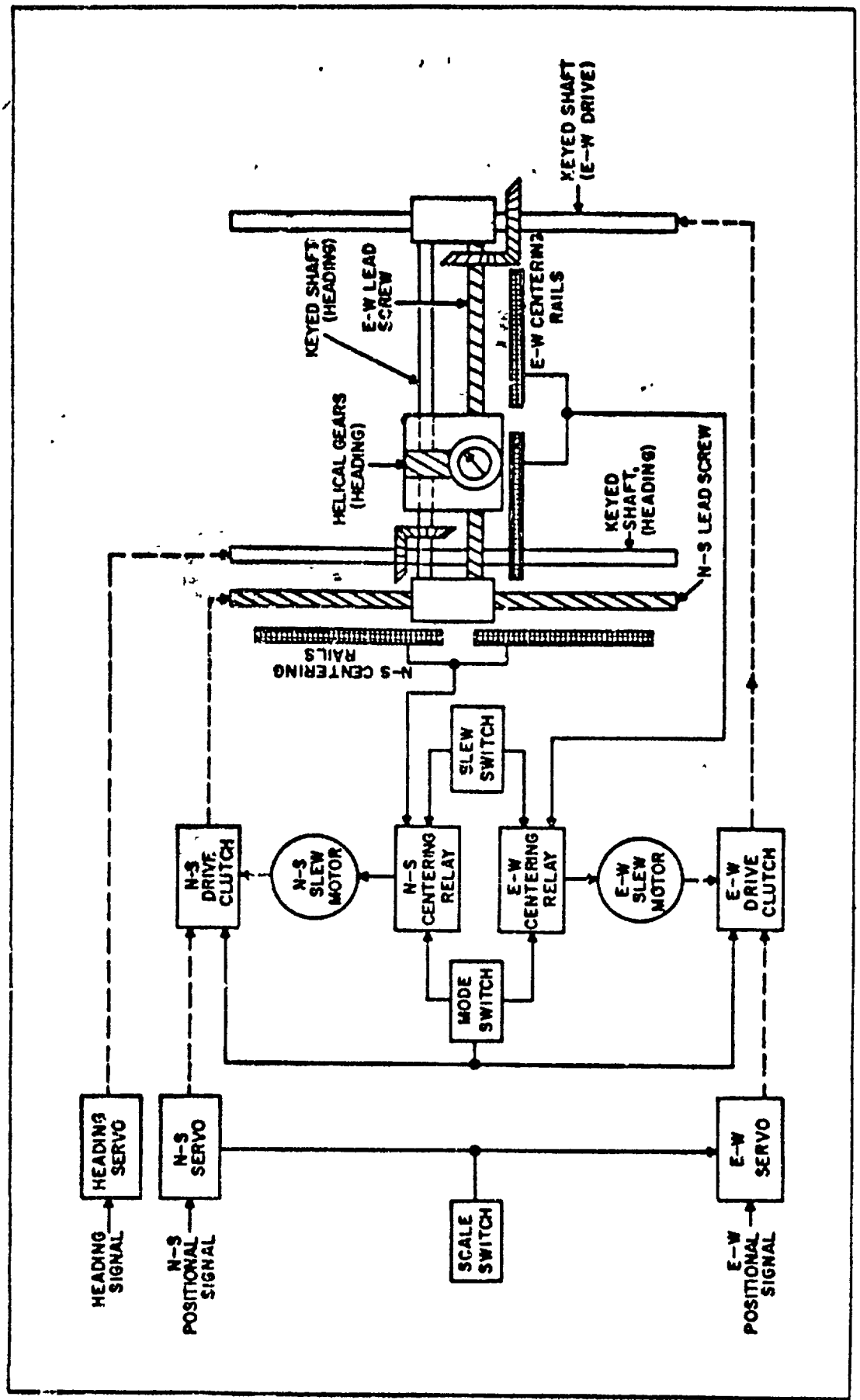



*Figure 2. Block Diagram, Plotting Board, Tactical Display PT-429/A*



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